The Setup (Our Simple Cloud)

- **Hardware:** We have 2 Virtual Machines (VM 1 and VM 2).
- **The Tasks:** The user submits 4 tasks in this exact order:
  - **Task A:** 100 Energy Units (Light)
  - **Task B:** 400 Energy Units (Heavy)
  - **Task C:** 50 Energy Units (Very Light)
  - **Task D:** 300 Energy Units (Heavy)

Logic 1: FCFS (First-Come, First-Served) - The Baseline

FCFS is "blind." It doesn't use your Machine Learning predictions. It just takes tasks in the exact order they arrive and hands them out alternately (Round-Robin style) to the next available machine.

1. Task A (100) arrives → Goes to **VM 1**.
2. Task B (400) arrives → Goes to **VM 2**.
3. Task C (50) arrives → Goes to **VM 1**.
4. Task D (300) arrives → Goes to **VM 2**.

The FCFS Result:

- **VM 1 Total Load:** 150 Units
- **VM 2 Total Load:** 700 Units (Severely Overloaded!)
- **System Peak Load: 700 Units** (This is our baseline penalty).

Logic 2: E-ACTS Optimized (Your Project)

Your system is "smart." It uses the ML predictions to know the Energy Units *before* execution, allowing it to sort and balance.

Step 1: Sort by Heaviest First
The system reorders the queue: Task B (400), Task D (300), Task A (100), Task C (50).

Step 2: Assign to the VM with the Lowest Current Load

1. Task B (400) → Both are at 0. Goes to **VM 1**. *(VM 1=400, VM 2=0)*
2. Task D (300) → VM 2 is lower (0 < 400). Goes to **VM 2**. *(VM 1=400, VM 2=300)*
3. Task A (100) → VM 2 is lower (300 < 400). Goes to **VM 2**. *(VM 1=400, VM 2=400)*
4. Task C (50) → Both are equal (400). Goes to **VM 1**. *(VM 1=450, VM 2=400)*

The Optimized Result:

- **VM 1 Total Load:** 450 Units
- **VM 2 Total Load:** 400 Units
- **System Peak Load: 450 Units** (Perfectly balanced!).

Logic 3: The Impact Analysis (How the UI gets its percentage)

Your dashboard compares the "Peak Load" of the dumb FCFS method against your smart Load Balanced method to show the user how much hardware stress/cooling energy they saved.

Here is the formula your backend will run to generate that final percentage on the screen:

$$Savings = \frac{FCFS_Peak - Optimized_Peak}{FCFS_Peak} \times 100$$

Plugging in our example numbers:

$$Savings = \frac{700 - 450}{700} \times 100$$

$$Savings = \frac{250}{700} \times 100$$

$$Savings = 35.7\%$$

What the UI displays: * Baseline (FCFS) Load: 700 Units

- E-ACTS Optimized Load: 450 Units
- **Projected Energy Savings: 35.7%**